SSC CGL 2024 Tier-I question paper with answers

13 September 2024 · Shift 2

74 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: +2 for a right answer, –0.50 for a wrong one

The other 26 of the paper's 100 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/Writeup_Final_Anwerkey_CGLE_2024_T1_191224.pdf

Compare your score: the official 2024 cut-off (Tier-I, all posts except JSO and SI Grade-II), out of 200

UR 153.19 **OBC** 146.26 **EWS** 142.02 **SC** 126.46 **ST** 111.89 **ESM** 69.93 **OH** 113.10 **HH** 64.79 **VH** 102.97 **Other PwD** 45.74

Cut-offs apply to marks normalised across all the exam's shifts, so a raw score on one shift is a rough guide. Source: the official result write-up, https://ssc.gov.in/api/attachment/uploads/masterData/NoticeBoards/Result%20Write-up%20CGLE%202024%20Tier%20I_51224.pdf

General Intelligence and Reasoning 17 questions

Q.1

In a certain code language, 'GLOBAL' is written as '221714272817', 'FOREST' is written as '23141124109', how will 'IMPACT' be written in that language?

(A) 20161328269
(B) 18141126247
(C) 20161226259
(D) 20161327259

Q.2

Select the triad in which the numbers are related to each other in the same way as the numbers in the following triads. 16-40-100 8-20-50 (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /deleting /multiplying etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 18-50-110
(B) 12-30-120
(C) 24-60-150
(D) 22-74-185

Q.5

Select the option that is related to the fifth term in the same way as the second term is related to the first term and the fourth term is related to the third term. READER : ERZWRE :: SERIAL : ESIRLA :: JINGLE : ?

(A) JIGNEL
(B) IJGNEL
(C) JIMTEL
(D) IJMTEL

Q.6

In a certain code language, ‘A + B’ means ‘A is the mother of B’, ‘A – B’ means ‘A is the father of B’, ‘A $ B’ means ‘A is the son of B’, ‘A % B’ means ‘A is the sister of B’, ‘A / B’ means ‘A is the wife of B’ and ‘A * B’ means ‘A is the husband of B’. Which of the following means P is the Son's wife of T? i. T / B – K * P + O ii. T + K * P + B % O iii. K * P + B % T + P iv. P + T * K + B % O

(A) Only ii
(B) iii and iv
(C) Only i
(D) Both i and ii

Q.7

In a certain code language, ‘RETAINED’ is written as ‘ENIATERD’ and ‘MATURITY’ is written as ‘TIRUTAMY’. How will ‘FIREWALL’ be written in that language?

(A) LAWEFIFR
(B) FLIREWAL
(C) LAWERIFL
(D) FLLAWERI

Q.8

What should come in place of the question mark (?) in the given series? 9 17 33 65 129 ?

(A) 259
(B) 257
(C) 258
(D) 254

Q.10

16 is related to 30 following a certain logic. Following the same logic, 81 is related to 110. To which of the following is 49 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /deleting /multiplying etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 60
(B) 64
(C) 90
(D) 72

Q.11

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some Toys are Plastic. Some Plastic are Metal. No Metal is Chemical. Conclusions: (I) Some Chemicals are Plastic. (II) No Toy is a Chemical.

(A) Only conclusion (I) follows.
(B) Both conclusions (I) and (II) follow.
(C) Neither conclusion (I) nor (II) follows.
(D) Only conclusion (II) follows.

Q.12

The position of how many letters will remain unchanged if each of the letters in the word ‘SIGHTED’ is arranged in the English alphabetical order?

(A) None
(B) Three
(C) Two
(D) One

Q.14

Which of the following terms will replace the question mark (?) in the given series? RLYV, PJWT, NHUR, ? , JDQN

(A) LFTP
(B) LGTP
(C) LFSP
(D) LGSP

Q.15

Three of the following four number-pairs are alike in a certain way and thus form a group. Which number-pair does NOT belong to that group? (NOTE: The relation should be found without breaking down the numbers into its constituent digits)

(A) 31, 41
(B) 73, 79
(C) 83, 97
(D) 61,71

Q.16

Read the given statements and conclusions carefully. You have to take the given statements to be true if they seem to be at variance from commonly known facts. You have to decide which conclusion(s) logically follow(s) from the given statements. Statements: All fruit are sour. Some sour are vegetables. Some vegetables are green. Conclusions: (I) All vegetables are sour. (II) Some sour are green.

(A) Neither conclusion (I) nor (II) follows
(B) Both conclusions (I) and (II) follow
(C) Only conclusion (II) follows
(D) Only conclusion (I) follows

Q.17

What will come in place of the question mark (?) in the following equation, if '×' is replaced by '÷', '÷' is replaced by '+', '+' is replaced by '×' and '–' remains unchanged? 42 × 3 ÷ 4 + 9 – 17 = ?

(A) 33
(B) 30
(C) 23
(D) 29

Q.21

What will come in place of the question mark (?) in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 107 + 2028 × 6 – 427 ÷ 7 = ?

(A) 2785
(B) 2758
(C) 2578
(D) 2268

Q.22

Select the option that represents the letters which, when sequentially placed from left to right in the blanks below, will complete the letter series. f h _ h _ g f _ h g _ h _ f f _ g _ h _ f

(A) g h f f g h f g
(B) g h f h g f f g
(C) g h f h g h f g
(D) g h f h g h h g

Q.23

Based on the position in the English alphabetical order, three of the following letter-clusters are alike in some manner and one is different. Select the odd letter-cluster. Note: The odd one out is not based on the number of consonants/vowels or their position in the letter cluster.

(A) SVXZ
(B) TWYZ
(C) EHJK
(D) YBDE

Q.24

Select the triad in which the numbers are related to each other in the same way as are the numbers of the given triads. (148, 126, 104), (98, 76, 54) (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /deleting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed)

(A) (149, 137, 115)
(B) (157, 135, 113)
(C) (128, 104, 82)
(D) (134, 102, 90)

General Awareness 24 questions

Q.1

Presently, which of the following countries is NOT a beneficiary of India's Duty-Free Tariff Preference (DFTP) Scheme?

(A) Rwanda
(B) Burundi
(C) Myanmar
(D) Angola

Q.2

A small object is placed on the focus on the left side of a convex lens. Where will be the image formed?

(A) At infinity on the right side of the lens.
(B) At infinity on the left side of the lens.
(C) At the focus on the right side of the lens.
(D) At the centre on the right side of the lens.

Q.3

After the fall of the Guptas, different kingdoms emerged in various parts of India. Among them, the Maukharis emerged in which of the following areas?

(A) Thanesar
(B) Valabhi
(C) Kunnur
(D) Kannauj

Q.4

Which of the following is a public sector industry?

(A) Adani
(B) BHEL
(C) Reliance
(D) Tata

Q.5

Indian National Army was formed in which of the following countries?

(A) Japan
(B) London
(C) Germany
(D) Singapore

Q.6

Which of the following nations won the ICC World Test Championship, defeating India, in 2023?

(A) Australia
(B) England
(C) South Africa
(D) New Zealand

Q.7

One of the greatest of Chola ruler Rajaraja I ruled from 985 AD to _____.

(A) 1015 AD
(B) 1017 AD
(C) 1018 AD
(D) 1014 AD

Q.8

Indian Olympic Association was established in which of the following years?

(A) 1929
(B) 1933
(C) 1927
(D) 1931

Q.9

Which of following is NOT a device of parliamentary proceedings?

(A) Half-an-hour discussion
(B) Full-day discussion
(C) Two-hour (short) discussion
(D) Zero-hour discussion

Q.10

Which of the following rivers marks the eastern-most boundary of the Himalayas?

(A) Kali
(B) Teesta
(C) Satluj
(D) Brahmaputra

Q.11

Padma Bhushan awardee, Ustad Sabri Khan played which of the following musical instruments?

(A) Tabla
(B) Sarod
(C) Sarangi
(D) Santoor

Q.12
Exfoliation is a form of _____.

(A) chemical weathering
(B) biochemical weathering
(C) physical weathering
(D) mass wasting

Q.13
Which of the following Articles deals with the composition of the Rajya Sabha?

(A) Article 92
(B) Article 76
(C) Article 84
(D) Article 80

Q.14
Which of the following is India's national academy of dance?

(A) Lalit Kala Akademi
(B) Centre for Indian Classical Dances
(C) Centre for Cultural Resources and Training
(D) Sangeet Natak Akademi

Q.15
What is the empirically fitted relationship between the rate of change of money, wage, and rate of unemployment known as?

(A) Keynesian model
(B) Philips curve
(C) Friedman's model
(D) Baumol hypothesis

Q.17
Who asked James Rennel to prepare the map of Hindustan?

(A) Lord Dalhousie
(B) Robert Clive
(C) Lord Ripen
(D) Warren Hastings

Q.18
The birth anniversary of Mahatma Gandhi is celebrated as the national festival of India. In which of the following years was Gandhi ji born?

(A) 1869
(B) 1890
(C) 1885
(D) 1877

Q.19
Who among the following is appointed as the new Chairman of Pension Fund Regulatory and Development Authority (PFRDA) in March 2023?

(A) Mamta Shankar
(B) Supratim Bandopadhyay
(C) Shasikanta Das
(D) Deepak Mohanty

Q.20
Transition elements are the elements that are found in Groups 3-12 of the modern periodic table, that constitute the _____.

(A) f-block
(B) d-block
(C) p-block
(D) s-block

Q.21
Which Article in the fundamental duty applies, when a construction worker takes his 8-year son to the site rather than sending him to school?

(A) Article 51A (h)
(B) Article 51A (i)
(C) Article 51A (j)
(D) Article 51A (k)

Q.22

The Nandi Award for Best Male Playback Singer was instituted by which state in the year 1977?

(A) Andhra Pradesh
(B) Maharashtra
(C) Tripura
(D) Gujarat

Q.23

Which Bill was passed by the Parliament, in July 2021, to help in the availability of working capital for the micro, small and medium enterprises sector?

(A) The Factoring Regulation (Amendment) Bill, 2021
(B) The Minerals Laws (Amendment) Bill, 2020
(C) The Insolvency and Bankruptcy Code (Amendment) Bill, 2021
(D) The Essential Commodities (Amendment) Bill, 2020

Q.24

Identify the structural formula for magnesium hydroxide.

(A) $Mg(OH)_2$
(B) MgO_2
(C) Mg_2H_2
(D) MgOH

Q.25

What kind of phyllotaxy was examined in the Guava plant?

(A) Superimposed
(B) Whorled
(C) Alternate
(D) Opposite

Quantitative Aptitude 9 questions

Q.2

P and Q are two fixed points 10 cm apart, and R is a point on PQ such that PR = 6 cm. By what percentage is the length of QR decreased when the length of PR is increased by 5%?

(A) 7%
(B) 8.5%
(C) 7.5%
(D) 8%

Q.3

A trader owes a merchant ₹9,810 due in 1 year, but the trader wants to settle the account after 6 months. If the rate of simple interest is 9% per annum, how much cash (in ₹) should he pay?

(A) 9,550
(B) 9,405
(C) 9,450
(D) 9,540

Q.4

A discount series of 10% and 16% on an invoice is the same as a single discount of:

(A) 24.4%
(B) 23.4%
(C) 21.4%
(D) 22.4%

Q.5

P and Q start running in opposite directions on a circular track from the same point. If their speeds are 10 m/s and 8 m/s, respectively, then after what time will they meet if the length of the track is 1620 m?

(A) 110 seconds
(B) 70 seconds
(C) 120 seconds
(D) 90 seconds

Q.6

Two different quantities of the same solution having ingredients A and B are stored in two different containers. In the first container, there are 252 litre of A and 441 litre of B. In the second container, the total quantity of the solution was 1188 litre. How much of the solution in the second container was made up of ingredient B?

(A) 756 litre
(B) 760 litre
(C) 765 litre
(D) 752 litre

Q.7

The radius of the circle is 8 cm. The distance of a point lying outside the circle from the centre is 17 cm. The length of the tangent drawn from the outside point to the circle is:

(A) 16 cm
(B) 19 cm
(C) 18 cm
(D) 15 cm

Q.8

Which digits should come in the place of x and y, respectively, if the number 62684xy is divisible by both 8 and 5?

(A) 0 and 5
(B) 4 and 0
(C) 5 and 0
(D) 2 and 0

Q.19

For which of the following values of a and b do the given equations, have NO solution? $x - ay = 2 - a$; $(1 - a)x + 6y = a + b$

(A) $a = 3, b \neq 1$
(B) $a = -3, b \neq 1$
(C) $a = -3, b \neq -1$
(D) $a = 3, b \neq -1$

Q.25

2 varieties of tea powders worth ₹250/kg and ₹300/kg are mixed with a third variety of tea powder in the ratio 1 : 1 : 2. If the mixture is worth ₹350/kg, find the price of the third variety tea powder.

(A) ₹500/kg
(B) ₹450/kg
(C) ₹400/kg
(D) ₹425/kg

English Comprehension 24 questions

Q.1

Select the option that expresses the given sentence in passive voice. The children have made the box.

(A) Children had made the box.
(B) The box is being made by the children.
(C) The box has been made by the children.
(D) The box had been made by the children.

Q.2

Select the option that can be used as a one-word substitute for the given group of words. One who eats a lot

(A) Chef
(B) Steward
(C) Chauffeur
(D) Glutton

Q.4

Select the most appropriate homophone to fill in the blank. The _____ was decorated with candles and chocolates everywhere.

(A) flor
(B) floor
(C) flour
(D) flower

Q.5

Select the option that can be used as a one-word substitute for the given group of words. A period of ten years

(A) Biennium
(B) Decade
(C) lustrum
(D) Century

Q.6

Select the option that can be used as a one-word substitute for the given group of words. a large, flat area surrounded by seats used for sports or entertainment

(A) Department
(B) Garden
(C) Senate
(D) Arena

Q.7

Parts of the following sentence have been given as options. Select the option that contains an error. I have the hundred reasons to not attend the meeting tomorrow.

(A) the meeting
(B) I have the hundred reasons
(C) to not attend
(D) tomorrow

Q.8

Given below are four jumbled sentences. Select the option that gives their correct logical sequence. A. When we connect this design with a wire, electricity flows. B. These metal strips were put in a sulphuric acid solution. C. He made two strips of different metals. D. Alessandro Volta discovered electric battery.

(A) CBDA
(B) DCBA
(C) BCDA
(D) ACDB

Q.9

Select the most appropriate meaning of the underlined idiom. All his friends admired him because he was a “straight shooter”.

(A) A thoroughly upright straightforward person
(B) Miserly and pessimistic individual
(C) Complicated and confusing person
(D) Excellent shooter

Q.10

Parts of a sentence are given below in jumbled order. Arrange the parts in the correct order to form a meaningful sentence. A. The upper part of the river B. Continuous rain on C. Week, there had been D. During the previous

(A) ADCB
(B) DBAC
(C) DCBA
(D) ABCD

Q.11

Select the option that expresses the given sentence in passive voice. They are organising a concert for charity.

(A) A concert is being organised by them for charity.
(B) They are organising a charity concert.
(C) A concert for charity is organising by them.
(D) They are being organised a concert for charity.

Q.12

Select the most appropriate ANTONYM of the underlined word. I was “happy” to know that Sheila passed her final exams.

(A) upbeat
(B) Sad
(C) Excited
(D) Joyful

Q.13

Select the most appropriate option to substitute the underlined word segment in the given sentence. The wedding was “put on” until January.

(A) put in
(B) put up
(C) put off
(D) put down

Q.14

Select the most appropriate ANTONYM of the word in brackets to fill in the blank. Sheila _____ (failed) all her classes.

(A) fizzled
(B) flopped
(C) flunked
(D) passed

Q.15

Select the most appropriate ANTONYM of the word in the brackets to fill in the blank. She _____ the target by an inch and lost the gold medal. [hit]

(A) cleaned
(B) missed
(C) bashed
(D) touched

Q.16

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. After I retire, I started a second career / as a writer and published several books / which allowed me to share my stories / and insights with a wider audience.

(A) which allowed me to share my stories
(B) After I retire, I started a second career
(C) and insights with a wider audience
(D) as a writer and published several books

Q.17

Select the INCORRECTLY spelt word.

(A) Occurence
(B) Harassment
(C) Supersede
(D) Exaggerate

Q.18

Select the most appropriate meaning of the underlined idiom. My uncle lives in New York, so I only see him “once in a blue moon”.

(A) Monthly
(B) Weekly
(C) Very often
(D) Not very often

Q.19

Select the INCORRECTLY spelt word

(A) Tangiable
(B) Terracotta
(C) Terracide
(D) Tertial

Q.20

Select the most appropriate synonym for the underlined word in the given sentence. One of Ram’s “leisure” activities is tennis.

(A) Languish
(B) Main
(C) Recreation
(D) Lead

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Earthquake is the (1)_____ breaking and displacement of substantial portions of the earth's outer rocky crust. During an earthquake, rock disturbances can cause rivers to change their flow. Landslides (2)_____ by earthquakes can cause significant destruction and loss of life. Tsunamis are a (3)_____ of damaging waves caused by large earthquakes that occur under the ocean. Earthquakes rarely directly kill anyone. Many (4)_____ and injuries are caused by falling debris and the fall of structures. The (5)_____ of an earthquake relies on how much and how far rock cracks and shifts.

Select the most appropriate option to fill in blank number 1.

(A) abrupt
(B) deliberate
(C) expansive
(D) innovative

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Earthquake is the (1)_____ breaking and displacement of substantial portions of the earth's outer rocky crust. During an earthquake, rock disturbances can cause rivers to change their flow. Landslides (2)_____ by earthquakes can cause significant destruction and loss of life. Tsunamis are a (3)_____ of damaging waves caused by large earthquakes that occur under the ocean. Earthquakes rarely directly kill anyone. Many (4)_____ and injuries are caused by falling debris and the fall of structures. The (5)_____ of an earthquake relies on how much and how far rock cracks and shifts.

Select the most appropriate option to fill in blank number 2.

(A) prevented
(B) blocked
(C) triggered
(D) subdued

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Earthquake is the (1)_____ breaking and displacement of substantial portions of the earth's outer rocky crust. During an earthquake, rock disturbances can cause rivers to change their flow. Landslides (2)_____ by earthquakes can cause significant destruction and loss of life. Tsunamis are a (3)_____ of damaging waves caused by large earthquakes that occur under the ocean. Earthquakes rarely directly kill anyone. Many (4)_____ and injuries are caused by falling debris and the fall of structures. The (5)_____ of an earthquake relies on how much and how far rock cracks and shifts.

Select the most appropriate option to fill in blank number 3.

(A) arrest
(B) interruption
(C) succession
(D) disorder

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Earthquake is the (1)_____ breaking and displacement of substantial portions of the earth's outer rocky crust. During an earthquake, rock disturbances can cause rivers to change their flow. Landslides (2)_____ by earthquakes can cause significant destruction and loss of life. Tsunamis are a (3)_____ of damaging waves caused by large earthquakes that occur under the ocean. Earthquakes rarely directly kill anyone. Many (4)_____ and injuries are caused by falling debris and the fall of structures. The (5)_____ of an earthquake relies on how much and how far rock cracks and shifts.

Select the most appropriate option to fill in blank number 4.

(A) viability
(B) survival
(C) fatalities
(D) durability

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Earthquake is the (1)_____ breaking and displacement of substantial portions of the earth's outer rocky crust. During an earthquake, rock disturbances can cause rivers to change their flow. Landslides (2)_____ by earthquakes can cause significant destruction and loss of life. Tsunamis are a (3)_____ of damaging waves caused by large earthquakes that occur under the ocean. Earthquakes rarely directly kill anyone. Many (4)_____ and injuries are caused by falling debris and the fall of structures. The (5)_____ of an earthquake relies on how much and how far rock cracks and shifts.

Select the most appropriate option to fill in blank number 5.

(A) depth

(B) magnitude

(C) corpse

(D) discipline

Answer key

General Intelligence and Reasoning

Q.1 **(A)**	Q.2 **(C)**	Q.5 **(D)**	Q.6 **(D)**	Q.7 **(C)**	Q.8 **(B)**
Q.10 **(D)**	Q.11 **(C)**	Q.12 **(C)**	Q.14 **(C)**	Q.15 **(B)**	Q.16 **(A)**
Q.17 **(A)**	Q.21 **(B)**	Q.22 **(D)**	Q.23 **(A)**	Q.24 **(B)**	

General Awareness

Q.1 **(D)**	Q.2 **(A)**	Q.3 **(D)**	Q.4 **(B)**	Q.5 **(D)**	Q.6 **(A)**
Q.7 **(D)**	Q.8 **(C)**	Q.9 **(B)**	Q.10 **(D)**	Q.11 **(C)**	Q.12 **(C)**
Q.13 **(D)**	Q.14 **(D)**	Q.15 **(B)**	Q.17 **(B)**	Q.18 **(A)**	Q.19 **(D)**
Q.20 **(B)**	Q.21 **(D)**	Q.22 **(A)**	Q.23 **(A)**	Q.24 **(A)**	Q.25 **(D)**

Quantitative Aptitude

Q.2 **(C)**	Q.3 **(B)**	Q.4 **(A)**	Q.5 **(D)**	Q.6 **(A)**	Q.7 **(D)**
Q.8 **(B)**	Q.19 **(D)**	Q.25 **(D)**			

English Comprehension

Q.1 **(C)**	Q.2 **(D)**	Q.4 **(B)**	Q.5 **(B)**	Q.6 **(D)**	Q.7 **(B)**
Q.8 **(B)**	Q.9 **(A)**	Q.10 **(C)**	Q.11 **(A)**	Q.12 **(B)**	Q.13 **(C)**
Q.14 **(D)**	Q.15 **(B)**	Q.16 **(B)**	Q.17 **(A)**	Q.18 **(D)**	Q.19 **(A)**
Q.20 **(C)**	Q.21 **(A)**	Q.22 **(C)**	Q.23 **(C)**	Q.24 **(C)**	Q.25 **(B)**

Explanations

General Intelligence and Reasoning

Q.1 · Answer (A) 20161328269

Each letter is coded by its position counted from Z (Z = 1, A = 26) plus 2: G = 20 + 2 = 22, L = 15 + 2 = 17 and so on. IMPACT: I = 18 + 2 = 20, M = 16, P = 13, A = 28, C = 26, T = 9, giving 20161328269.

Q.2 · Answer (C) 24-60-150

Each number is 2.5 times the one before: 16 → 40 → 100 and 8 → 20 → 50. Likewise 24 → 60 → 150.

Q.5 · Answer (D) IJMTEL

The first two letters swap, the middle two are replaced by their opposite letters (A↔Z, D↔W, R↔I) and the last two swap: READER → ER ZW RE and SERIAL → ES IR LA. So JINGLE → IJ MT EL: IJMTEL.

Q.6 · Answer (D) Both i and ii

In i, T is B's wife and B is K's father, so T is K's mother, and K is P's husband: P is T's son's wife. In ii, T is K's mother and K is P's husband directly. iii makes P and T each the other's mother, and in iv P is T's mother.

Q.7 · Answer (C) LAWERIFL

The first seven letters are reversed and the eighth stays put: RETAINE + D → ENIATER + D and MATURIT + Y → TIRUTAM + Y. So FIREWAL + L → LAWERIF + L: LAWERIFL.

Q.8 · Answer (B) 257

Each term is twice the one before minus 1: 9 × 2 − 1 = 17, 17 × 2 − 1 = 33 and so on, so 129 × 2 − 1 = 257.

Q.10 · Answer (D) 72

16 = 4^2 and 30 = 5 × 6; 81 = 9^2 and 110 = 10 × 11. So 49 = 7^2 goes with 8 × 9 = 72.

Q.11 · Answer (C) Neither conclusion (I) nor (II) follows.

Metal is kept apart from chemicals, but nothing fixes how chemicals relate to plastic or to toys. Neither 'some chemicals are plastic' (I) nor 'no toy is a chemical' (II) is certain.

Q.12 · Answer (C) Two

In alphabetical order SIGHTED becomes D E G H I S T. Comparing position by position, G (3rd) and H (4th) stay where they were: two letters.

Q.14 · Answer (C) LFSP

Each letter moves 2 places back each time: R, P, N, L, J; L, J, H, F, D; Y, W, U, S, Q; V, T, R, P, N. The missing cluster is LFSP.

Q.15 · Answer (B) 73, 79

All eight numbers are primes. 31 and 41 have one prime between them (37), as do 83 and 97 (89) and 61 and 71 (67). 73 and 79 are consecutive primes, with none between, so they do not belong.

Q.16 · Answer (A) Neither conclusion (I) nor (II) follows

'Some sour are vegetables' does not make all vegetables sour (I), and the green vegetables need not be among the sour ones (II). Neither follows.

Q.17 · Answer (A) 33

After replacing the signs: 42 ÷ 3 + 4 × 9 − 17 = 14 + 36 − 17 = 33.

Q.21 · Answer (B) 2758

After interchanging the signs: 107 − 2028 ÷ 6 + 427 × 7 = 107 − 338 + 2989 = 2758.

Q.22 · Answer (D) g h f h g h h g

With g, h, f, h, g, h, h, g in the blanks the block fhghhgf repeats: fhghhgf fhghhgf fhghhgf.

Q.23 · Answer (A) SVXZ

TWYZ, EHJK and YBDE move +3, +2, +1 (Y to B wraps round). SVXZ moves +3, +2, +2, so it is the odd one.

Q.24 · Answer (B) (157, 135, 113)
In each triad the numbers fall by 22: 148, 126, 104 and 98, 76, 54. Likewise 157, 135, 113.

General Awareness

Q.1 · Answer (D) Angola
SSC's key gives Angola. India's DFTP scheme is for the least developed countries (LDCs), and Angola left the UN's list of LDCs in February 2021; Rwanda, Burundi and Myanmar are still on it.

Q.2 · Answer (A) At infinity on the right side of the lens.
Rays from an object at the principal focus of a convex lens leave it parallel, so the image forms at infinity on the other side.

Q.3 · Answer (D) Kannauj
The Maukharis ruled from Kannauj after the Guptas declined; Thanesar was the seat of the Pushyabhutis and Valabhi of the Maitrakas.

Q.4 · Answer (B) BHEL
Bharat Heavy Electricals Limited (BHEL) is owned by the Government of India; Adani, Reliance and Tata are private groups.

Q.5 · Answer (D) Singapore
The Indian National Army was organised in Singapore in 1942 from Indian prisoners of war, and Subhas Chandra Bose took command of it there in 1943.

Q.6 · Answer (A) Australia
Australia beat India by 209 runs in the 2023 World Test Championship final at the Oval.

Q.7 · Answer (D) 1014 AD
Rajaraja I ruled from 985 to 1014 AD; his son Rajendra I succeeded him.

Q.8 · Answer (C) 1927
The Indian Olympic Association was set up in 1927.

Q.9 · Answer (B) Full-day discussion
Half-an-hour discussions, short-duration (two-hour) discussions and Zero Hour are all devices of parliamentary proceedings; there is no 'full-day discussion'.

Q.10 · Answer (D) Brahmaputra
In NCERT's division of the Himalayas by river valleys, the Brahmaputra marks their eastern-most boundary; the Satluj, Kali and Teesta divide sections within the range.

Q.11 · Answer (C) Sarangi
Ustad Sabri Khan was one of India's best-known sarangi players.

Q.12 · Answer (C) physical weathering
Exfoliation, the peeling away of rock in curved sheets, is caused by physical forces such as expansion and contraction with temperature changes, so it is physical weathering.

Q.13 · Answer (D) Article 80
Article 80 sets out the composition of the Council of States (Rajya Sabha); Article 81 does the same for the Lok Sabha.

Q.14 · Answer (D) Sangeet Natak Akademi
The Sangeet Natak Akademi is India's national academy for music, dance and drama; the Lalit Kala Akademi is for the visual arts.

Q.15 · Answer (B) Philips curve
A. W. Phillips fitted an inverse relationship between the rate of change of money wages and the rate of unemployment, from British data: the Phillips curve.

Q.17 · Answer (B) Robert Clive
Robert Clive asked James Rennell, the East India Company's surveyor in Bengal, to produce maps of Hindustan.

Q.18 · Answer (A) 1869
Mahatma Gandhi was born on 2 October 1869 at Porbandar; Gandhi Jayanti marks the day.

Q.19 · Answer (D) Deepak Mohanty
Deepak Mohanty, a former Executive Director of the Reserve Bank of India, was appointed Chairman of the PFRDA in March 2023.

Q.20 · Answer (B) d-block
Groups 3 to 12, in which the d orbitals are being filled, make up the d-block: the transition elements.

Q.21 · Answer (D) Article 51A (k)
Article 51A(k), added by the 86th Amendment in 2002, makes it the duty of a parent or guardian to provide opportunities for education to a child between six and fourteen.

Q.22 · Answer (A) Andhra Pradesh
SSC's key gives Andhra Pradesh, whose government presents the Nandi Awards for Telugu cinema.

Q.23 · Answer (A) The Factoring Regulation (Amendment) Bill, 2021
The Factoring Regulation (Amendment) Bill, 2021 let many more NBFCs take up factoring (buying firms' receivables), easing working capital for MSMEs.

Q.24 · Answer (A) $Mg(OH)_2$
Magnesium forms Mg^{2+} ions, each balanced by two hydroxide ions (OH^-): $Mg(OH)_2$.

Q.25 · Answer (D) Opposite
NCERT gives guava, with Calotropis, as an example of opposite phyllotaxy: a pair of leaves at each node, facing each other.

Quantitative Aptitude

Q.2 · Answer (C) 7.5%
QR is 10 – 6 = 4 cm. PR rises 5% to 6.3 cm, so QR falls to 3.7 cm: a decrease of 0.3/4 = 7.5%.

Q.3 · Answer (B) 9,405
SSC's key values the debt from today: ₹9,810/1.09 = ₹9,000 now, which grows to 9,000 × 1.045 = ₹9,405 in six months at 9% a year. (Discounting ₹9,810 straight back six months, 9,810/1.045 ≈ ₹9,387.56, is not among the options.)

Q.4 · Answer (A) 24.4%
After 10% and then 16% off, 0.9 × 0.84 = 0.756 of the price is left: a single discount of 24.4%.

Q.5 · Answer (D) 90 seconds
Running in opposite directions they close the gap at 10 + 8 = 18 m/s, so they meet after 1620/18 = 90 seconds.

Q.6 · Answer (A) 756 litre
It is the same solution, so A : B is the same in both containers: 252 : 441 = 4 : 7. B in the second = 1188 × 7/11 = 756 litres.

Q.7 · Answer (D) 15 cm
The tangent is perpendicular to the radius at the point of contact: $\sqrt{(17^2 - 8^2)} = \sqrt{(289 - 64)} = \sqrt{225} = 15$ cm.

Q.8 · Answer (B) 4 and 0
Divisible by both 5 and 8, the number must end in 0, so y = 0. The last three digits, 4x0, must then be divisible by 8: of the options only 440 is (8 × 55), so x = 4.

Q.19 · Answer (D) a = 3,b ≠-1

No solution needs $1/(1 - a) = -a/6 \neq (2 - a)/(a + b)$. The equality gives $a^2 - a - 6 = 0$, so a = 3 or a = –2. With a = 3 the first two ratios are –1/2, and $-1/(3 + b) \neq -1/2$ needs b ≠ –1.

Q.25 · Answer (D) ₹425/kg

$(250 + 300 + 2x)/4 = 350$, so 550 + 2x = 1,400 and x = ₹425/kg.

English Comprehension

Q.1 · Answer (C) The box has been made by the children.

The present perfect 'have made' becomes 'has been made': 'The box has been made by the children.'

Q.2 · Answer (D) Glutton

A glutton eats far too much; a chef cooks, a steward serves and a chauffeur drives.

Q.4 · Answer (B) floor

Only 'floor', the surface of a room, fits the sentence; 'flour' and 'flower' sound alike but mean something else.

Q.5 · Answer (B) Decade

A decade is ten years; a biennium is two, a lustrum five and a century a hundred.

Q.6 · Answer (D) Arena

An arena is a large, level area surrounded by seats for sports or shows.

Q.7 · Answer (B) I have the hundred reasons

An unspecified number takes the indefinite article: 'I have a hundred reasons', not 'the hundred reasons'.

Q.8 · Answer (B) DCBA

D introduces Volta's discovery, C says how he began (two strips of different metals), B what he did with them and A the result: DCBA.

Q.9 · Answer (A) A thoroughly upright straightforward person

A 'straight shooter' is an honest, straightforward person.

Q.10 · Answer (C) DCBA

'During the previous (D) week, there had been (C) continuous rain on (B) the upper part of the river (A).'

Q.11 · Answer (A) A concert is being organised by them for charity.

The present continuous 'are organising' becomes 'is being organised': 'A concert is being organised by them for charity.'

Q.12 · Answer (B) Sad

The opposite of 'happy' is 'sad'.

Q.13 · Answer (C) put off

To 'put off' is to postpone: the wedding was postponed until January. To 'put on' is to wear or stage something.

Q.14 · Answer (D) passed

The opposite of 'failed' is 'passed'; 'fizzled', 'flopped' and 'flunked' all mean failed.

Q.15 · Answer (B) missed

The opposite of 'hit' is 'missed', which is why she lost the gold medal.

Q.16 · Answer (B) After I retire, I started a second career

The sentence is in the past ('started', 'published'), so 'After I retired', not 'After I retire'.

Q.17 · Answer (A) Occurence

The correct spelling is 'occurrence'.

Q.18 · Answer (D) Not very often

'Once in a blue moon' means very rarely.

Q.19 · Answer (A) Tangiable
The correct spelling is 'tangible'.

Q.20 · Answer (C) Recreation
'Leisure' activities are those done in one's free time, for 'recreation'.

Q.21 · Answer (A) abrupt
An earthquake is the 'abrupt' (sudden) breaking and displacement of the crust.

Q.22 · Answer (C) triggered
Landslides 'triggered' (set off) by earthquakes.

Q.23 · Answer (C) succession
A tsunami is a 'succession' (series) of damaging waves.

Q.24 · Answer (C) fatalities
Earthquakes rarely kill directly, but falling debris and structures cause many 'fatalities' (deaths) and injuries.

Q.25 · Answer (B) magnitude
The 'magnitude' (size) of an earthquake depends on how much and how far the rock cracks and shifts.

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